Create a knowledge base using propositional logic and show that the given query entails the knowledge base or not

```
# Truth Table Enumeration for Propositional Entailment

from itertools import product

def implies(a, b):
    return (not a) or b

# Knowledge Base
def KB(P, Q, R):
    s1 = implies(Q, P)
    s2 = implies(P, not Q)
    s3 = Q or R
    return s1, s2, s3, s1 and s2 and s3

# Queries
def query_R(P, Q, R):
    return R

def query_RP(P, Q, R):
    return implies(R, P)

def query_QR(P, Q, R):
    return implies(Q, R)

# Display truth table
print("\nTRUTH TABLE")
print("-" * 95)

print(f"{'P':^5}{'Q':^5}{'R':^5}"
      f"{'Q->P':^8}{'P->~Q':^8}{'Q|R':^8}"
      f"{'KB':^8}{'R':^8}{'R->P':^8}{'Q->R':^8}")

print("-" * 95)

models = []

for P, Q, R in product([False, True], repeat=3):

    s1, s2, s3, kb = KB(P, Q, R)

    qr = query_R(P, Q, R)
    rp = query_RP(P, Q, R)
    qqr = query_QR(P, Q, R)
```

```
    models.append((P, Q, R, kb, qr, rp, qqr))

    values = [P, Q, R, s1, s2, s3, kb, qr, rp, qqr]

    print("".join(f"{str(v):^8}" for v in values))

print("-" * 95)

# Check entailment
def entails(query_index):
    for model in models:
        if model[3] and not model[query_index]:
            return False
    return True

print("\nENTAILMENT RESULTS")
print("-" * 40)

print("KB entails R      :", entails(4))
print("KB entails R -> P :", entails(5))
print("KB entails Q -> R :", entails(6))

print("-" * 40)
```

OUTPUT :

```
TRUTH TABLE
-----------------------------------------------------------------------------------------
  P    Q    R    Q->P   P->~Q    Q|R      KB      R      R->P    Q->R
-----------------------------------------------------------------------------------------
 False  False  False   True    True   False  False  False   True    True
 False  False   True    True    True    True   True   True  False   True
 False   True  False  False    True    True  False  False   True  False
 False   True   True  False    True    True  False   True  False   True
  True  False  False   True    True  False  False  False   True    True
  True  False   True    True    True    True   True   True   True    True
  True   True  False   True   False    True  False  False   True  False
  True   True   True    True   False    True  False   True   True    True
-----------------------------------------------------------------------------------------

ENTAILMENT RESULTS
----------------------------------------
KB entails R      : True
KB entails R -> P : False
KB entails Q -> R : True
```